Magitech

Machinery built to hold a working, and workings designed around what a machine can do that a hand cannot. Both halves are old; putting them together deliberately is not, and the people doing it have not agreed on what to call it. **Magitech** is the word that has stuck outside the trade, mostly because it is short.

Two ends of the same problem

Melisor Magocracy came at it from the magic. An academy that can already build a teleportation circle wants to know what a circle can be made to do when it is a component rather than a destination, and Melisor has the theorists, the ley-lines and the funding to ask.

Silicar came at it from the machine. A guild that has spent four generations on locks, gearing and irrigation wants to know which of its mechanisms get better with a working inside them, and Silicar has the workshops, the clockwork trade and a habit of building the thing before arguing about it. It is also the guild that invented the firelock, by accident, and has never once let that trade and this one share a bench. See Technology.

Neither of them finds this funny. Everybody else does.

Materials and power

A magitech device is built, not grown out of a spellbook, and it looks it. The good ones use metals and woods that got rare the moment the trade started buying them up, finished the way a luthier finishes a violin rather than the way a smith finishes a plough, and set with a gemstone or crystal more often than not. The metal itself is usually an alloy — a single pure metal holds a working's shape badly, and every workshop guards its own blend like a vintner guards a cellar.

Power is the harder problem, and neither academy solved it from nothing. An aeon stone in this trade is one gem carrying two separate cuts, not a battery wired to a separate working. The **charge rune** comes first and pulls raw magical energy through the gem, and the gem is not an interchangeable vessel for that: some stones catalyse the draw cleanly and some waste half of what the rune is trying to pull through them, which is why a workshop pays for a good ruby and not merely a red one. The **property rune** comes second, into the same stone, and shapes whatever the charge rune drew — and it doesn't take to every gem the same way either. Fire cuts cleanly into ruby and badly into almost anything else; a ward or open air takes to diamond; earth wants peridot. Cut the right rune into the wrong stone and the device is weak or doesn't hold at all, which is most of why the trade's pairings look fixed rather than arbitrary. Both cuts are made when the stone is made and neither can be reworked after. Nobody reprograms an aeon stone; they requisition another, in the right gem this time, and cut it again. Twenty years ago nobody in either academy could cut either rune, let alone say which stone suited which. What they had were whole low-level stones pulled out of old ruins, enough of them, studied long enough, to work both cuts back out from the finished article — and to notice, eventually, that the ruins had never once wasted a ruby on anything but fire.

The same ruins gave up broken housings too: alloy and wood still carrying a third cut, **circuitry**, carved into the material itself rather than the stone, that once carried a stone's shaped power the rest of the way into whatever the housing was built to do. That is where the trade got its wiring without inventing it a third time.

Property runes keep turning up in new varieties, and the trade keeps copying or inventing past them — fire, earth, air, a ward, a solvent, a season, with no sign the list is finished. The charge rune never has. Every stone anyone has ever pulled from a ruin carries the same basic cut, and in twenty years nobody has managed to design, or find, a more advanced one. A better stone or a cleaner alloy can close the gap between a device and its theoretical best, but neither can raise the ceiling itself. That ceiling is one specific cut that nobody has learned to improve. Whether that is a limit of what survived the ruins or a limit neither academy has cracked on its own, Melisor and Silicar give two different answers in public and privately admit to the same one: they don't know.

Not everything that ships is a find. A few pieces are close reconstructions of something pulled half-whole out of a ruin and finished in a workshop; the rest are new work, designed from scratch around the recovered runework and the recovered circuitry, that happened to solve a problem the empire had already solved once. Nobody building either kind knows which one they're holding, because nobody building either kind knows there is a difference to ask about.

The list below names each device by its property rune only — the charge rune riding under it goes unmentioned the way nobody points out the nails in a chair. Twenty years in, Silicar's workshops have shipped a handful of things:

- **The wayfinder** — a disc small enough to wear on a cord, pointing toward whatever its stone was keyed to find. The one every court now wants and almost none can buy; Silicar's guild fills perhaps thirty orders a year.
- **The cold lamp** — no flame, no oil, no working laid on fresh each evening. A stone behind hammered glass, burning level and cold until it runs dry.
- **The strike-stone** — a ruby cut with a fire rune, set into a case of alloy the size of a thumb. Click it and it lights. No oil, no flint, no wick, which makes it the one magitech item an ordinary household might actually own, on the rare year it can find one to buy.
- **The slate** — a peridot cut with an earth rune, bedded into a tablet of worked stone. Write on it, press the rune, and the mark is gone — a scribe's whole complaint against parchment, solved. Thurion Merchant Alliance's counting-houses are already trying to buy the patent out from under Silicar before a rival house does.
- **The diving mask** — a diamond cut with an air rune, set into the glass. A salvage crew wearing one can work a wreck for the better part of an hour on a single charge, which has already changed what a reef-city considers worth the dive.

Every one of them is rare, every one is expensive, and every one works exactly as many times as its stones allow, and not once more.

Prestige pieces

Melisor Magocracy never needed magitech to solve anything. An academy that can put three teleportation circles under one campus has never once lacked a practical answer, so its half of the trade was never aimed at a problem. It was aimed at whoever already has every option and wants to be envied for the one they picked.

- **The warded clasp** — a diamond cut with an abjuration rune, pinned at the collar. It raises the same half-shield a first-year apprentice learns to cast, and holds it just as well, which has made it wildly popular with nobles who want to duel without earning the privilege the ordinary way. Ashwin Devaraj, who has spent a career being paid for that privilege and cited for none of it, has opinions about the clasp and states them loudly, usually in a tavern.
- **The presser** — a topaz cut with a cleansing rune, set into a flat iron. A launderer does the same work for a fraction of the price and cannot be shown off at a dinner party, which is rather the point of owning one instead.
- **The seasoning dish** — a citrine cut with a transmutation rune, laid under the centrepiece at a Council banquet. No two academies season a dish the same way, and a Thelemar dinner argues its politics through the courses as much as through anyone's conversation.

None of it is cheaper than paying somebody to do the job. That was never the offer, and neither was making it last: a warded clasp cut at Thelemar runs dry in a fraction of the uses a Silicar lamp gets out of the same grade of stone, because the circuitry under it was carved by somebody who has never had to care. Nobody on the Council has ever once complained.

How it is reckoned

Every bench in Brightfurrow has a shorthand chalked over it, three quantities weighed against each other — what the stone draws, how well its rune suits the gem, what the circuit does to it on the way out — and most of the people who use it daily could not derive it. Neither academy can. Between them they hold most of the pieces and neither holds a theory, because each has spent twenty years building out the half it was already good at.

Silicar has the numbers. Four generations of guild practice, written down the way a guild writes things down: tables. Which alloy carries a fire working and which sulks at it. How deep to cut a track before the gain stops paying for the brittleness. What a peridot costs to seat badly. The Brightfurrow guilds can hand an apprentice a book that will let him build a better lamp than anyone in Thelemar, and not one line of that book explains why any of it is true. Ask a guildmaster what impedance *is* and you will get an alloy recipe, correctly, and a look. Their weakness is their strength worn the other way round. They cannot predict a pairing they have not already tried, so every new working costs them a season of ruined stones to find the material for.

Melisor has the grammar. Runes are arcane theory and arcane theory is Thelemar's own ground, so the academies can do the thing Silicar cannot: read a property rune, classify it, and cut a new one on purpose. That is where the widening list of workings comes from — fire, earth, air, a ward, a solvent, a season — and Melisor can usually say in advance which gem a new rune will want, which no guild table can. What no academy has is any account of the housing whatsoever. Circuitry is craft, craft is a trade for people who work with their hands, and a chair who took an interest would be asked about it at dinner. So Thelemar cuts superb stones and seats them in circuits a Brightfurrow journeyman would be ashamed of, and never connects that to why its clasps run dry.

What neither of them has is the part that only appears when the two halves are set down together — that the stone, the cut and the housing are three terms of one relation, and that the relation has laws. Getting there needs somebody who has stood at both benches, which neither academy will permit and both would regard as a waste of a career. See Thaumodynamics.

The stakes

- **It is the first field in a century where Melisor Magocracy is not obviously ahead.** Silicar is small, is not a great power, and is producing results the academies did not predict.
- **The Council of Archmages cannot rule on it**, because two academies claim the same process and the Council's horror of ruling on anything is a policy older than the dispute.
- **Everything it produces is unlicensed**, in the same way the firelock is unlicensed. See Technology.
- **It is expensive and it is visible.** A magitech workshop is a large building with a great deal of money in it and no guild old enough to have arranged its own protection.

You cannot repeat any of this. Not to the party, not on paper, not in stone.

Thaumodynamics

TOWERS OF SAEROTH · PLAYER HANDOUT

Twenty years of Magitech and nobody has written down what the trade is actually doing. Melisor Magocracy has theorists who will not go near a bench and Silicar has benchwrights who will not go near a theory, and between the two of them the field has accumulated a great deal of practice and no account of itself. What follows is an attempt at one. It is not published, it would embarrass both academies if it were, and the name for the study is the author's own coinage.

The fundamental bench equation

$$\mathbf{W = Q \cdot P \div Z}$$

Chalked over every bench in Brightfurrow, usually by somebody who could not derive it. It estimates the useful Working a single-stone apparatus produces across one full cycle:

Term	Name	What it is
W	Working	The useful magical effect the apparatus actually delivers
Q	Charge	Thaumic potential the stone's charge rune makes available
P	Pattern	How well the property rune shapes that charge into the effect wanted
Z	Impedance	The opposition the material circuit puts up as patterned charge passes through it

So $\mathbf{W = f(Q, P, Z)}$, and each of the three has a law governing it.

Charge is set by the charge rune, which fixes the ceiling, and by the gem, which decides how near the ceiling a given stone comes: $\mathbf{0 \le Q \le Q_{max}}$. Every charge rune anybody has recovered gives the same Qmax, which is the whole of why the trade is stuck where it is.

Pattern operates on whatever charge is available, giving the product $\mathbf{Q \cdot P}$. It is an effectiveness, not an identity — *which* property a rune expresses comes from the rune, and P says only how much of that property survives the cutting.

Impedance depends on the housing rather than the stone: $\mathbf{Z = f(M, C, \tau)}$, for material, circuit geometry, and the kind of Working being carried. The three are set out under the Third Law below.

Zeroth Law: Thaumic Affinity

> A pattern expresses itself only in proportion to its affinity with the stone it is cut into.

$$\mathbf{0 \le P \le P_{max}}$$

It is numbered zeroth because the other three laws all use P and none of them can say what P is. A property rune is not equally effective in every gem; its geometry has to agree with the crystal it is cut into. Fire agrees with ruby. A ward, or open air, agrees with diamond. Earth wants peridot. Fire cut into a gem with no affinity for it gives a weak effect or an unstable one, and a perfect pairing approaches Pmax without ever passing it.

This is why particular stones became associated with particular workings, and why that association is not a convention anybody chose.

First Law: Conservation of Charge

> A stone gives up no more charge than it was given. How fast it is spent can be changed. The total cannot.

$$\int_0^T (-dQ/dt)\,dt \le Q(0)$$

where **t** is time, **T** the moment the working ends, and **Q(0)** the charge in the stone when it began.

A firelock empties a stone over a fraction of a second. A lamp takes months about it. Both are bounded by the same integral, and this is the formal statement of what the trade already says in plainer words: a device works as many times as its stone allows and not once more.

Read what this law does not say. It bounds the charge a stone can surrender. It says nothing about the Working that charge produces, and those are not the same quantity. Conflating them is the mistake that cost the author two seasons and is corrected under the Third Law below.

Second Law: Thaumic Transfer

> What arrives as Working is what the pattern shaped and the circuit carried, less everything spilled on the way and plus whatever the circuit was good enough to draw in alongside it.

Charge leaves the stone at a rate **$-(dQ/dt)$**. Some of it arrives as the effect the device was built for. Some is spilled. And a circuit carved well enough couples the channel to the same ambient field the charge rune pulls from, so that a third quantity arrives which the stone never supplied:

$$-(dQ/dt) + a(t) = w(t) + \ell(t)$$

Term	Name	What it is
w(t)	Working rate	Arriving, at that instant, as the effect the device was built for
ℓ(t)	Loss	Spent at that instant and arriving as anything else
a(t)	Ambient draw	Field the circuit couples in on its own account, which the stone never held
η(t)	Transfer	The net of all three, as a multiple of what the stone gave up

Net it out and the ratio of Pattern to Impedance is what has been measuring the whole business all along:

$\eta(t) = P(t) \div Z(t)$, and **$w(t) = \eta(t) \cdot (-dQ/dt)$**

$\eta < 1$** is a circuit losing more than it couples, which is nearly every circuit anybody currently makes. **$\eta > 1$ is a circuit that gives back more than the stone handed it, and it is not getting that from nowhere — it is getting it from the field, along the channel the property rune opened, and it cannot open that channel without the stone's flow to hold it open.

Waste leaves as heat, light, vibration, backlash, degradation of the stone itself, or an effect nobody asked for. A badly made lighter that draws ten units of charge to deliver six has not lost the other four. It has put them somewhere, and a workshop that cannot say where has a fire hazard rather than a device. This is the whole reason good circuitry makes a stone last: not because it draws less, but because less of what it draws goes astray.

Third Law: Impedance

> Every material circuit opposes, modifies, and — carved well enough — amplifies what passes through it. The quality of a Working therefore depends on the material and the geometry together, and no circuit can be driven past what its own material will bear.

$Z > 0$, always, and **$Z = f(M, C, \tau)$**.

Term	Name	What it is
M	Material	What the housing is made of: the alloy blend or the wood, its purity, its grain
C	Circuit	The geometry of the carving — its path, corners, depth and workmanship
τ	Working type	Which kind of Working is being carried, a circuit fit for one being no guarantee for another

Impedance is not a property of a substance the way hardness is. Two devices cut from identical steel can differ wildly because one engineer carved a far better circuit, and a superb circuit carved into the wrong material still performs badly. The τ term matters as much: the same circuit that carries a ward cleanly may be poor at carrying fire, so impedance has to be quoted against the working it is meant to carry or it means nothing.

None of the three is fixed while a device runs, either. Heat changes what a material will carry, and a circuit that cracks is a circuit recut by accident, so a Z measured on a cold bench is a Z measured under conditions the device will not stay in.

Impedance below Pattern is gain. Drive Z under P and the device returns more Working than the stone gave up, which sounds like the First Law being broken and is not: the surplus is ambient field, and the stone is still emptying at exactly the rate the First Law allows. Charge is conserved. Working was never the conserved quantity, and taking it for one is an error this treatise made for two seasons and has had to unpick.

But every circuit has a floor it cannot be driven under. Call it **Z_{min}**, and understand it as a breaking point rather than a limit approached:

$$Z > Z_{min}(M, C)$$

Below that value the circuit is holding a coupling wider than its own material can carry, and it does not quietly stop improving. It fails. The carving cracks along its own tracks, or the stone goes at once instead of over months, or the working arrives somewhere nobody aimed it. A device built at gain is a device built nearer its own destruction, and the better the material the nearer you may safely go.

Zmin is not one number. It belongs to the material and the carving together, it moves with heat and with damage, and nobody has mapped it for so much as a single alloy. It is the open question of the field, and the reason the answer matters is that every unit of gain is worth more than a better gem.

What the laws already answer

Three things fall out that the bench does not have the vocabulary to notice.

The bench equation is not a simplification. It is exact, under conditions worth naming. Hold P and Z steady through the cycle and run the stone to exhaustion, and the integral collapses:

$$W = \int_0^T \eta \cdot (-dQ/dt)\, dt = (P \div Z) \cdot [Q(0) - Q(T)] = Q(0) \cdot P \div Z$$

So the chalked version is the full theory in the case a bench usually cares about, with **Q** meaning **Q(0)**, a fresh stone. It fails in two situations and both are worth knowing. On a partly spent stone the bracket is **Q(0) − Q(T)** and not Q(0). And on a device whose circuit heats or damages as it runs, Z drifts mid-working and cannot be pulled out of the integral — which means the shorthand is least reliable exactly where impedance is already worst. A workshop that trusts the chalk is trusting it hardest on the devices it should trust it least on.

An earlier draft of this treatise put the floor at Zmin = 1, on the reasoning that a circuit returning more than it was handed would be an apparatus outrunning its own stone, which conservation forbids. The reasoning was wrong and the error is worth naming, because it is the error the whole trade makes. Conservation binds *charge*. A circuit at gain does not refill the stone, does not slow its emptying, and does not violate anything: it opens a wider channel onto the field and the stone pays for holding it open. Two seasons went into a bound that was never there.

What is there instead is **Zmin(M, C)**, a breaking point rather than a ceiling, different for every material and carving, and unmapped. That is a worse answer and a truer one.

Which leaves two roads rather than one. Everything a device finally delivers comes out of what the stone was handed, **Qmax**, multiplied by what the circuit does with it. A better gem closes the gap to Qmax. A better circuit multiplies whatever crosses it, and there is no reason in any of the four laws why that multiplier stops at one.

Twenty years of craft has produced no device more powerful than the first ones out of the ground, and the trade reads that as proof it is stuck behind the charge rune. It is not proof of anything of the kind. Present circuits sit a little under unity — twenty years has carried the field from badly lossy to nearly break-even, which is real work and is not the same as gain. The road past Qmax was never closed. Nobody has walked it, because nobody has written down that it is there.

You cannot repeat any of this. Not to the party, not on paper, not in stone.